图像处理：第五次作业

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1 Z 变换性质

(a) 若 $x(n)$ 的 Z 变换为 $X(z)$，则 $(-1)^n x(n)$ 的 Z 变换为 $X(-z)$

由于 $x(n)$ 的 Z 变换为:

$$X(z) = \sum_{-\infty}^{\infty} x(n) z^{-n}$$

因为对称性质:

$$(-1)^n = (-1)^{-n}$$

所以 $(-1)^n x(n)$ 的 Z 变换为:

$$\sum_{-\infty}^{\infty} (-1)^n x(n) z^{-n} = \sum_{-\infty}^{\infty} (-1)^{-n} x(n) z^{-n} = \sum_{-\infty}^{\infty} x(n)(-z)^{-n} = X(-z)$$

(b) 若 $x(n)$ 的 Z 变换为 $X(z)$，则 $x(-n)$ 的 Z 变换为 $X\left(\frac{1}{z}\right)$

由于:

$$X(z) = \sum_{-\infty}^{\infty} x(n) z^{-n}$$

令 $k = -n$，则:

$$\sum_{-\infty}^{\infty} x(-n) z^{-n} = \sum_{-\infty}^{\infty} x(k) z^{k}$$

因此:

$$\sum_{-\infty}^{\infty} x(k) z^{k} = \sum_{-\infty}^{\infty} x(k)(z^{-1})^{-k} = X\left(\frac{1}{z}\right)$$

(c) 若 $x(n)$ 的 Z 变换为 $X(z)$，则有信号“抽取”在 Z 域的性质

若 $x(n)$ 的 Z 变换为 $X(z)$，则 $x_{\text{down}}(n) = x(2n) \Leftrightarrow X_{\text{down}}(z) = \frac{1}{2}\left[X(\sqrt{z}) + X(-\sqrt{z})\right]$

$$X_{\text{down}}(z) = \sum_{-\infty}^{\infty} x_{\text{down}}(n) z^{-n} = \sum_{-\infty}^{\infty} x(2n) z^{-n}$$

由于:

$$X(\sqrt{z}) = \sum_{-\infty}^{\infty} x(k)(\sqrt{z})^{-k} = \sum_{-\infty}^{\infty} x(k)z^{-k/2}$$
$$X(-\sqrt{z}) = \sum_{-\infty}^{\infty} x(k)(-\sqrt{z})^{-k} = \sum_{-\infty}^{\infty} x(k)(-1)^{-k}z^{-k/2} = \sum_{-\infty}^{\infty} x(k)(-1)^{k}z^{-k/2}$$

叠加后得到:

$$\begin{aligned} X(\sqrt{z}) + X(-\sqrt{z}) &= \sum_{-\infty}^{\infty} x(k)z^{-k/2} + \sum_{-\infty}^{\infty} x(k)(-1)^{k}z^{-k/2} \\ &= \sum_{-\infty}^{\infty} x(k)\left[1 + (-1)^{k}\right] z^{-k/2} \end{aligned}$$

当 k 为偶数，$1 + (-1)^k = 2$；当 k 为奇数，$1 + (-1)^k = 0$，所以:

$$X(\sqrt{z}) + X(-\sqrt{z}) = \sum_{k\in\text{even}} x(k) \cdot 2 \cdot z^{-k/2}$$

令 $k = 2m$，其中 $m \in \mathbb{Z}$，则:

$$X(\sqrt{z}) + X(-\sqrt{z}) = 2 \cdot \sum_{-\infty}^{\infty} x(2m)z^{-m}$$

因此:

$$X_{\text{down}}(z) = \sum_{-\infty}^{\infty} x(2n)z^{-n} = \frac{1}{2}\left[2 \cdot \sum_{-\infty}^{\infty} x(2m)z^{-m}\right] = \frac{1}{2}\left[X(\sqrt{z}) + X(-\sqrt{z})\right]$$

2 正交镜像滤波器组推论

$$G_1(z) = -z^{-2k+1}G_0(-z^{-1}) \implies g_1(n) = (-1)^n g_0(2k-1-n)$$

由于:

$$G_1(z) = -z^{-2k+1}G_0(-z^{-1}) = -z^{-2k+1}\sum_{-\infty}^{\infty} g_0(n)(-z^{-1})^{-n} = \sum_{-\infty}^{\infty} g_0(n)(-1)^{n+1}z^{n+1-2k}$$

令 $-m = n + 1 - 2k$，则:

$$\sum_{-\infty}^{\infty} g_0(n)(-1)^{n+1}z^{n+1-2k} = \sum_{-\infty}^{\infty} g_0(2k-1-m)(-1)^{2k-m}z^{-m}$$

又因为:

$$(-1)^{2k-m}=(-1)^{2k}\cdot(-1)^{-m}=1\cdot(-1)^{m}=(-1)^{m}$$

所以:

$$\sum_{-\infty}^{\infty}g_0(2k-1-m)(-1)^{2k-m}z^{-m}=\sum_{-\infty}^{\infty}g_0(2k-1-m)(-1)^{m}z^{-m}$$

因此:

$$g_1(n)=(-1)^n g_0(2k-1-n)$$

3 正交族小波变换完美重建滤波器组推论

由于高通和低通综合滤波器交替反转关系:

$$g_1(n)=(-1)^n g_0(2k-1-n)$$

又因为分析滤波器是综合滤波器时域镜像正交关系:

$$\begin{cases} h_0(n)=g_0(2k-1-n) \\ h_1(n)=g_1(2k-1-n) \end{cases}$$

反转表达:

$$h_0(n)=g_0(2k-1-n)\implies h_0(2k-1-n)=g_0(n)$$

由:

$$h_1(n)=g_1(2k-1-n)=(-1)^{2k-1-n}g_0\left[2k-1-(2k-1-n)\right]=(-1)^{n+1}g_0(n)$$

即:

$$h_1(n)=(-1)^{n+1}h_0(2k-1-n)$$

4 哈尔小波变换矩阵

哈尔基函数:

$$h_k(z) = h_{pq}(z) = \frac{1}{\sqrt{N}} \begin{cases} 2^{p/2}, & \dfrac{q-1}{2^p} \leqslant z < \dfrac{q-0.5}{2^p} \\ -2^{p/2}, & \dfrac{q-0.5}{2^p} \leqslant z < \dfrac{q}{2^p} \\ 0, & z \in \left[0,\ \dfrac{q-1}{2^p}\right) \bigcup \left[\dfrac{q}{2^p},\ 1\right] \end{cases}$$

根据公式 $k = 2^p + q - 1$，可得:

表 1: 哈尔变换参数组

k	p	q
0	0	0
1	0	1
2	1	1
3	1	2
4	2	1
5	2	2
6	2	3
7	2	4
8	3	1
9	3	2
10	3	3
11	3	4
12	3	5
13	3	6
14	3	7
15	3	8

以 $k = 6$ 为例，能满足条件的最大 2^p 是 4，所以 $p = 2$，因此 $q = k - 2^p + 1 = 3$，根据 $h_6(z)$ 哈尔基函数:

$$h_6(z) = \frac{1}{\sqrt{16}} \begin{cases} 2^{2/2}, & \dfrac{3-1}{2^2} \leqslant z < \dfrac{3-0.5}{2^2} \\ -2^{2/2}, & \dfrac{3-0.5}{2^2} \leqslant z < \dfrac{3}{2^2} \\ 0, & z \in \left[0,\ \dfrac{3-1}{2^2}\right) \bigcup \left[\dfrac{3}{2^2},\ 1\right] \end{cases} = \begin{cases} 2, & 0.5 \leqslant z < 0.625 \\ -2, & 0.625 \leqslant z < 0.75 \\ 0, & z \in [0,\ 0.5) \bigcup [0.75,\ 1] \end{cases}$$

采样点 z 的取值即填充列，包含 $\dfrac{0}{16}$，$\dfrac{1}{16}$，$\dfrac{2}{16}$，$\ldots$，$\dfrac{15}{16}$ 共 16 个值，因为 $\dfrac{8}{16} = 0.5$，$\dfrac{9}{16} =$

0.5625, $\dfrac{10}{16}=0.625$, $\dfrac{11}{16}=0.6875$，所以哈尔变换矩阵第 7 行，即 $h_6(z)$ 的 16 个元素为：

$$[0,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0,\ \frac{2}{4},\ \frac{2}{4}\ -\frac{2}{4},\ -\frac{2}{4},\ 0,\ 0,\ 0,\ 0]$$

以此类推，可得 H_{16} 矩阵：

$$H_{16}=\frac{1}{4}\begin{bmatrix}
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \\
\sqrt{2} & \sqrt{2} & \sqrt{2} & \sqrt{2} & -\sqrt{2} & -\sqrt{2} & -\sqrt{2} & -\sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sqrt{2} & \sqrt{2} & \sqrt{2} & \sqrt{2} & -\sqrt{2} & -\sqrt{2} & -\sqrt{2} & -\sqrt{2} \\
2 & 2 & -2 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 2 & 2 & -2 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 2 & -2 & -2 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 2 & -2 & -2 \\
2\sqrt{2} & -2\sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 2\sqrt{2} & -2\sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 2\sqrt{2} & -2\sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 2\sqrt{2} & -2\sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2\sqrt{2} & -2\sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2\sqrt{2} & -2\sqrt{2} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2\sqrt{2} & -2\sqrt{2} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2\sqrt{2} & -2\sqrt{2}
\end{bmatrix}$$

5 二元组扩展与其系数

(a) $\varphi_0=[1/\sqrt{2},\ 1/\sqrt{2}]^\top$, $\varphi_1=[1/\sqrt{2},\ -1/\sqrt{2}]^\top$

$$\alpha_0=\varphi_0^\top\cdot[3,\ 2]^\top=[1/\sqrt{2},\ 1/\sqrt{2}]\cdot\begin{bmatrix}3\\2\end{bmatrix}=\frac{5\sqrt{2}}{2}$$

$$\alpha_1=\varphi_1^\top\cdot[3,\ 2]^\top=[1/\sqrt{2},\ -1/\sqrt{2}]\cdot\begin{bmatrix}3\\2\end{bmatrix}=\frac{\sqrt{2}}{2}$$

$$\alpha_0\varphi_0+\alpha_1\varphi_1=\frac{5\sqrt{2}}{2}\begin{bmatrix}\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}\end{bmatrix}+\frac{\sqrt{2}}{2}\begin{bmatrix}\frac{1}{\sqrt{2}}\\ -\frac{1}{\sqrt{2}}\end{bmatrix}=[3,\ 2]^\top$$

(b) $\varphi_0=[1,\ 0]^\top$, $\varphi_1=[1,\ 1]^\top$**，基础的对偶** $\overline{\varphi}_0=[1,\ -1]^\top$, $\overline{\varphi}_1=[0,\ 1]^\top$

$$\alpha_0=\overline{\varphi}_0^\top\cdot[3,\ 2]^\top=[1,\ -1]\cdot\begin{bmatrix}3\\2\end{bmatrix}=1$$

$$\alpha_1=\overline{\varphi}_1^\top\cdot[3,\ 2]^\top=[0,\ 1]\cdot\begin{bmatrix}3\\2\end{bmatrix}=2$$

$$\alpha_0\varphi_0+\alpha_1\varphi_1=1\times\begin{bmatrix}1\\ \\0\end{bmatrix}+2\begin{bmatrix}1\\ \\1\end{bmatrix}=[3,\ 2]^\top$$

(c) $\varphi_0=[1,\ 0]^\top,\ \varphi_1=[-1/2,\ \sqrt{3}/2]^\top,\ \varphi_2=[-1/2,\ -\sqrt{3}/2]^\top,\ \overline{\varphi}_i=2\varphi_i/3$

$$\alpha_0=\overline{\varphi}_0^\top\cdot[3,\ 2]^\top=\begin{bmatrix}\frac{2}{3},\ 0\end{bmatrix}\cdot\begin{bmatrix}3\\2\end{bmatrix}=2$$

$$\alpha_1=\overline{\varphi}_1^\top\cdot[3,\ 2]^\top=\begin{bmatrix}-\frac{1}{3},\ \frac{\sqrt{3}}{3}\end{bmatrix}\cdot\begin{bmatrix}3\\2\end{bmatrix}=-1+\frac{2\sqrt{3}}{3}$$

$$\alpha_2=\overline{\varphi}_2^\top\cdot[3,\ 2]^\top=\begin{bmatrix}-\frac{1}{3},\ -\frac{\sqrt{3}}{3}\end{bmatrix}\cdot\begin{bmatrix}3\\2\end{bmatrix}=-1-\frac{2\sqrt{3}}{3}$$

$$\alpha_0\varphi_0+\alpha_1\varphi_1+\alpha_2\varphi_2=2\times\begin{bmatrix}1\\ \\0\end{bmatrix}+\left(\frac{2\sqrt{3}}{3}-1\right)\begin{bmatrix}-\frac{1}{2}\\ \\ \frac{\sqrt{3}}{2}\end{bmatrix}-\left(\frac{2\sqrt{3}}{3}+1\right)\begin{bmatrix}-\frac{1}{2}\\ \\ -\frac{\sqrt{3}}{2}\end{bmatrix}=[3,\ 2]^\top$$

6 尺度函数未满足多分辨率分析的第二要求

$$\varphi(x)=\begin{cases}1 & 0.25\leqslant x<0.75\\ \\ 0 & \text{other}\end{cases}$$

W_j 空间的所有 $k\in\mathbb{Z}$ 定义小波集合 $\{\psi_{j,k}(x)\}$：

$$\psi_{j,k}(x)=2^{j/2}\psi(2^jx-k)$$

因此：

$$\begin{aligned}\varphi_{0,0}(x)&=\varphi(x)\\ \varphi_{1,0}(x)&=\sqrt{2}\varphi(2x)\\ \varphi_{1,1}(x)&=\sqrt{2}\varphi(2x-1)\end{aligned}$$

因为 $\varphi_{0,0}(x)$ 的支撑集是 $[0.25,\ 0.75)$，$\varphi_{1,0}(x)$ 的支撑集是 $[0.125,\ 0.375)$，$\varphi_{1,1}(x)$ 的支撑集是 $[0.625,\ 0.875)$，所以区间 $[0.375,\ 0.625)$ 出现空洞，即 $\varphi_{1,0}(x)=\varphi_{1,1}(x)\equiv 0$，而 $\varphi_{0,0}(x)\equiv 1$，因此意味着无论这两个函数如何加权，都无法填满中间这段空隙，即 $\varphi_{0,0}(x)$ 无法由 $\varphi_{1,0}(x)$ 和 $\varphi_{1,1}(x)$ 加权求和表示。

7 离散小波变换

DWT 起始尺度 j 的函数:

$$W_\varphi(j_0,\ k) = \frac{1}{\sqrt{M}} \sum_x f(x)\varphi_{j_0,k}(x)$$
$$W_\psi(j,\ k) = \frac{1}{\sqrt{M}} \sum_x f(x)\psi_{j,k}(x)$$

$$f(0) = 1 \qquad f(1) = 4 \qquad f(2) = -3 \qquad f(3) = 0$$

哈尔尺度函数:

$$\varphi(x) = \begin{cases} 1 & 0 \leqslant x < 1 \\ \\ 0 & \text{other} \end{cases}$$

哈尔小波函数:

$$\psi(x) = \begin{cases} 1 & 0 \leqslant x < 0.5 \\ \\ -1 & \quad 0.5 \leqslant x < 1 \\ \\ 0 & \text{other} \end{cases}$$

小波集合:

$$\psi_{j,k}(x) = 2^{j/2}\psi\left(2^j \frac{x}{M} - k\right)$$

(a) 计算一维 DWT

若 $j_0 = 1$:

$$\begin{aligned} W_\varphi(1,0) &= \frac{1}{\sqrt{4}}\left[f(0)\varphi_{1,0}(0) + f(1)\varphi_{1,0}(1) + f(2)\varphi_{1,0}(2) + f(3)\varphi_{1,0}(3)\right] \\ &= \frac{1}{2}\left(1 \times \sqrt{2} + 4 \times \sqrt{2} - 3 \times 0 + 0 \times 0\right) \\ &= \frac{5\sqrt{2}}{2} \end{aligned}$$

$$
\begin{aligned}
W_\varphi(1,1) &= \frac{1}{\sqrt{4}}\left[f(0)\varphi_{1,1}(0)+f(1)\varphi_{1,1}(1)+f(2)\varphi_{1,1}(2)+f(3)\varphi_{1,1}(3)\right] \\
&= \frac{1}{2}\left(1\times 0+4\times 0-3\times\sqrt{2}+0\times\sqrt{2}\right) \\
&= -\frac{3\sqrt{2}}{2}
\end{aligned}
$$

$$
\begin{aligned}
W_\psi(1,0) &= \frac{1}{\sqrt{4}}\left[f(0)\psi_{1,0}(0)+f(1)\psi_{1,0}(1)+f(2)\psi_{1,0}(2)+f(3)\psi_{1,0}(3)\right] \\
&= \frac{1}{2}\left[1\times\sqrt{2}+4\times\left(-\sqrt{2}\right)-3\times 0+0\times 0\right] \\
&= -\frac{3\sqrt{2}}{2}
\end{aligned}
$$

$$
\begin{aligned}
W_\psi(1,1) &= \frac{1}{\sqrt{4}}\left[f(0)\psi_{1,1}(0)+f(1)\psi_{1,1}(1)+f(2)\psi_{1,1}(2)+f(3)\psi_{1,1}(3)\right] \\
&= \frac{1}{2}\left[1\times 0+4\times 0-3\times\sqrt{2}+0\times\left(-\sqrt{2}\right)\right] \\
&= -\frac{3\sqrt{2}}{2}
\end{aligned}
$$

因此，DWT 集合是 $\left\{\frac{5\sqrt{2}}{2},\ -\frac{3\sqrt{2}}{2},\ -\frac{3\sqrt{2}}{2},\ -\frac{3\sqrt{2}}{2}\right\}$。

(b) 根据变换值计算 $f(1)$

根据:

$$
f(x)=\frac{1}{\sqrt{M}}\sum_k W_\varphi(j_0,k)\varphi_{j_0,k}(x)+\frac{1}{\sqrt{M}}\sum_{j=j_0}^{\infty}\sum_k W_\psi(j,k)\psi_{j,k}(x)
$$

则有:

$$
\begin{aligned}
f(x) &= \frac{1}{\sqrt{4}}\left[W_\varphi(1,0)\varphi_{1,0}(x)+W_\varphi(1,1)\varphi_{1,1}(x)+W_\psi(1,0)\psi_{1,0}(x)+W_\psi(1,1)\psi_{1,1}(x)\right] \\
&= \frac{1}{2}\left[\frac{5\sqrt{2}}{2}\varphi_{1,0}(x)-\frac{3\sqrt{2}}{2}\varphi_{1,1}(x)-\frac{3\sqrt{2}}{2}\psi_{1,0}(x)-\frac{3\sqrt{2}}{2}\psi_{1,1}(x)\right]
\end{aligned}
$$

因此:

$$
f(1)=\frac{1}{2}\times\left(\frac{5\sqrt{2}}{2}\times\sqrt{2}-\frac{3\sqrt{2}}{2}\times 0-\frac{3\sqrt{2}}{2}\times\sqrt{2}-\frac{3\sqrt{2}}{2}\times 0\right)=1
$$

8 完美重建来此小波变换分解后的滤波器输出

信号 $f = [1,\ 3,\ 5,\ 7,\ 4,\ 3,\ 2,\ 1]$，利用哈尔小波进行两层快速小波变换分解，计算各层的滤波器输出，然后再进行完美重建。其中:

$$\text{高通滤波器 } h_1 = \left[-\frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}}\right]$$
$$\text{低通滤波器 } h_0 = \left[\frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}}\right]$$

高通卷积:

$$f \circledast h_1 = \left[-\frac{1}{\sqrt{2}},\ -\frac{2}{\sqrt{2}},\ -\frac{2}{\sqrt{2}},\ -\frac{2}{\sqrt{2}},\ \frac{3}{\sqrt{2}},\ \frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}}\right]$$

低通卷积:

$$f \circledast h_0 = \left[\frac{1}{\sqrt{2}},\ \frac{4}{\sqrt{2}},\ \frac{8}{\sqrt{2}},\ \frac{12}{\sqrt{2}},\ \frac{11}{\sqrt{2}},\ \frac{7}{\sqrt{2}},\ \frac{5}{\sqrt{2}},\ \frac{3}{\sqrt{2}},\ \frac{1}{\sqrt{2}}\right]$$

由于 $2\downarrow$ 每隔一个点取一个值，所以:

$$W_\psi(1,n) = \left\{-\frac{2}{\sqrt{2}},\ -\frac{2}{\sqrt{2}},\ \frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}}\right\}$$
$$W_\varphi(1,n) = \left\{\frac{4}{\sqrt{2}},\ \frac{12}{\sqrt{2}},\ \frac{7}{\sqrt{2}},\ \frac{3}{\sqrt{2}}\right\}$$

上支路卷积:

$$W_\varphi(1,n) \circledast h_1 = \left[-2,\ -4,\ \frac{5}{2},\ 2,\ \frac{3}{2}\right]$$

下支路卷积:

$$W_\varphi(1,n) \circledast h_0 = \left[2,\ 8,\ \frac{19}{2},\ 5,\ \frac{3}{2}\right]$$

由于 $2\downarrow$ 每隔一个点取一个值，所以第二级:

$$W_\psi(0,n) = \{-4,\ 2\}$$
$$W_\varphi(0,n) = \{8,\ 5\}$$

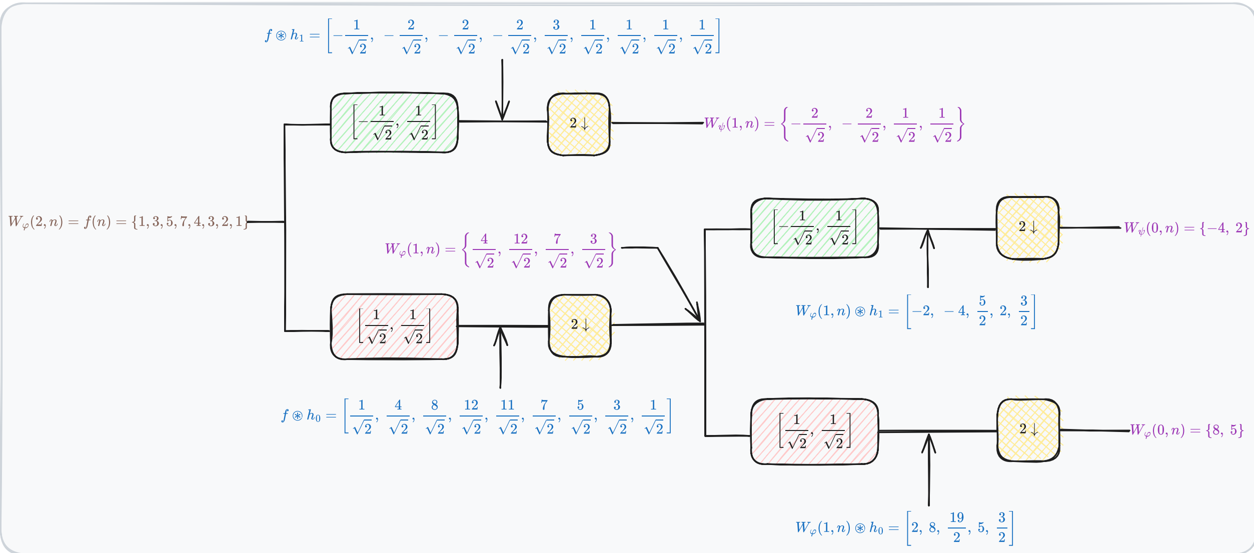

图 1: 用哈尔尺度和小波向量计算序列 f 的二尺度快速小波变换

$$\text{高通综合滤波器 } h_1 = \left[\frac{1}{\sqrt{2}},\ -\frac{1}{\sqrt{2}}\right]$$
$$\text{低通综合滤波器 } h_0 = \left[\frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}}\right]$$

由于 $2\uparrow$ 每隔一个点补充零值，所以:

$$W_\psi(0,n) = \{-4,\ 2\} \to \{-4,\ 0,\ 2,\ 0\}$$
$$W_\varphi(0,n) = \{8,\ 5\} \to \{8,\ 0,\ 5,\ 0\}$$

高通综合卷积:

$$\{-4,\ 0,\ 2,\ 0\} \circledast h_1 = \left[-\frac{4}{\sqrt{2}},\ \frac{4}{\sqrt{2}},\ \frac{2}{\sqrt{2}},\ -\frac{2}{\sqrt{2}},\ 0\right]$$

低通综合卷积:

$$\{8,\ 0,\ 5,\ 0\} \circledast h_0 = \left[\frac{8}{\sqrt{2}},\ \frac{8}{\sqrt{2}},\ \frac{5}{\sqrt{2}},\ \frac{5}{\sqrt{2}},\ 0\right]$$

叠加:

$$\begin{aligned} W_\varphi(1,n) &= \left[-\frac{4}{\sqrt{2}},\ \frac{4}{\sqrt{2}},\ \frac{2}{\sqrt{2}},\ -\frac{2}{\sqrt{2}},\ 0\right] + \left[\frac{8}{\sqrt{2}},\ \frac{8}{\sqrt{2}},\ \frac{5}{\sqrt{2}},\ \frac{5}{\sqrt{2}},\ 0\right] \\ &= \left[\frac{4}{\sqrt{2}},\ \frac{12}{\sqrt{2}},\ \frac{7}{\sqrt{2}},\ \frac{3}{\sqrt{2}},\ 0\right] \end{aligned}$$

由于 2↑ 每隔一个点补充零值，所以第二级：

$$\left[\frac{4}{\sqrt{2}},\ \frac{12}{\sqrt{2}},\ \frac{7}{\sqrt{2}},\ \frac{3}{\sqrt{2}},\ 0\right] \to \left[\frac{4}{\sqrt{2}},\ 0,\ \frac{12}{\sqrt{2}},\ 0,\ \frac{7}{\sqrt{2}},\ 0,\ \frac{3}{\sqrt{2}}\right]$$

第二级低通综合卷积：

$$\left[\frac{4}{\sqrt{2}},\ 0,\ \frac{12}{\sqrt{2}},\ 0,\ \frac{7}{\sqrt{2}},\ 0,\ \frac{3}{\sqrt{2}}\right] \circledast h_0 = \left[2,\ 2,\ 6,\ 6,\ \frac{7}{2},\ \frac{7}{2},\ \frac{3}{2},\ \frac{3}{2}\right]$$

又因为：

$$W_\psi(1,n) = \left\{-\frac{2}{\sqrt{2}},\ -\frac{2}{\sqrt{2}},\ \frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}},\ 0\right\}$$

所以 2↑ 每隔一个点补充零值，得到：

$$\left[-\frac{2}{\sqrt{2}},\ -\frac{2}{\sqrt{2}},\ \frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}},\ 0\right] \to \left[-\frac{2}{\sqrt{2}},\ 0,\ -\frac{2}{\sqrt{2}},\ 0,\ \frac{1}{\sqrt{2}},\ 0,\ \frac{1}{\sqrt{2}}\right]$$

第二级高通综合卷积：

$$\left[-\frac{2}{\sqrt{2}},\ 0,\ -\frac{2}{\sqrt{2}},\ 0,\ \frac{1}{\sqrt{2}},\ 0,\ \frac{1}{\sqrt{2}}\right] \circledast h_1 = \left[-1,\ 1,\ -1,\ 1,\ \frac{1}{2},\ -\frac{1}{2},\ \frac{1}{2},\ -\frac{1}{2}\right]$$

第二级叠加：

$$\begin{aligned} W_\varphi(2,n) = f(n) &= \left[-1,\ 1,\ -1,\ 1,\ \frac{1}{2},\ -\frac{1}{2},\ \frac{1}{2},\ -\frac{1}{2}\right] + \left[2,\ 2,\ 6,\ 6,\ \frac{7}{2},\ \frac{7}{2},\ \frac{3}{2},\ \frac{3}{2}\right] \\ &= [1,\ 3,\ 5,\ 7,\ 4,\ 3,\ 2,\ 1] \end{aligned}$$

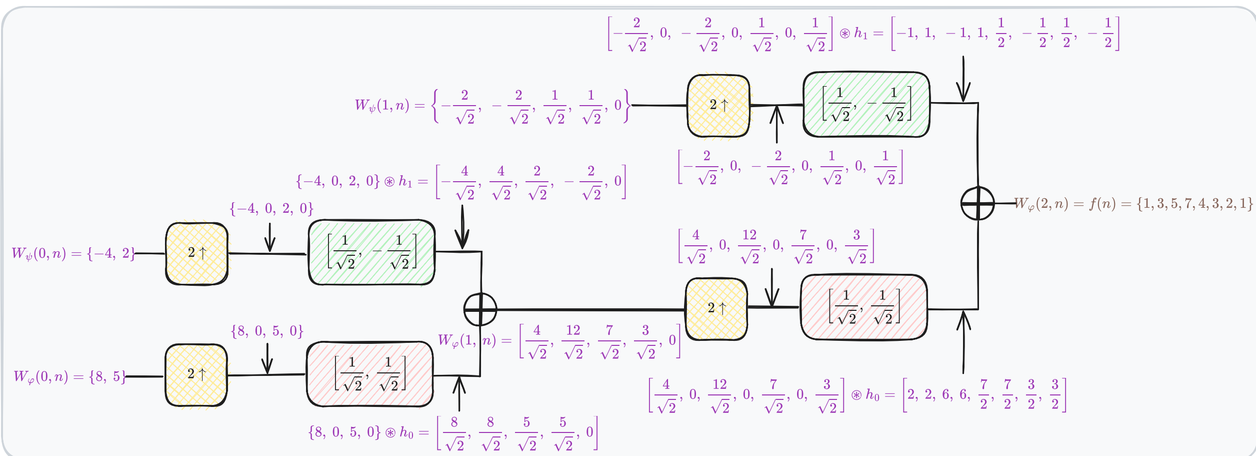

图 2: 用哈尔尺度和小波向量计算序列 $W_\psi(0,n), W_\varphi(0,n), W_\psi(1,n)$ 的二尺度快速小波逆变换